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REVIEW

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The functional and molecular effects of problematic alcohol consumption on skeletal muscle: a focus on athletic performance

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ABSTRACT

Background: Chronic alcohol misuse is associated with alcoholic myopathy, characterized by skeletal muscle weakness and atrophy. Moreover, there is evidence that sports-related people seem to exhibit a greater prevalence of problematic alcohol consumption, especially binge drinking (BD), which might not cause alcoholic myopathy but can negatively impact muscle function and amateur and professional athletic performance.

Objective: To review the literature concerning the effects of alcohol consumption on skeletal muscle function and structure that can affect muscle performance.

Methodology: We examined the currently available literature (PubMed, Google Scholars) to develop a narrative review summarizing the knowledge about the effects of alcohol on skeletal muscle function and exercise performance, obtained from studies in human beings and animal models for problematic alcohol consumption.

Results: Exercise- and sport-based studies indicate that alcohol consumption can negatively affect muscle recovery after vigorous exercise, especially in men, while women seem less affected. Clinical studies and pre-clinical laboratory research have led to the knowledge of some of the mechanisms involved in alcohol-related muscle dysfunction, including an imbalance between anabolic and catabolic pathways, reduced regeneration, increased inflammation and fibrosis, and deficiencies in energetic balance and mitochondrial function. These pathological features can appear not only under chronic alcohol misuse but also in other alcohol consumption patterns.

Conclusions: Most laboratory-based studies use chronic or acute alcohol exposure, while episodic BD, the most common drinking pattern in amateur and professional athletes, is underrepresented. Nevertheless, alcohol consumption negatively affects skeletal muscle health through different mechanisms, which collectively might contribute to reduced sports performance.

RESUMEN

Introduccion: El consumo crónico del alcohol se ha asociado al desarrollo de la miopatía alcohólica, una condición del músculo esquelético que se caracteriza por debilitamiento muscular y atrofia. Por otro lado, la evidencia sugiere que existe una alta prevalencia del consumo problemático de alcohol en deportistas, lo que ocurre principalmente en un patrón por atracón (Binge Drinking, BD, en ingles) y que puede impactar negativamente la funcion muscular, y por lo tanto, el desempeño deportivo.

Objetivo: Revisar en la literatura los efectos del consumo del alcohol en la estructura y función del músculo esquelético que podrían afectar el rendimiento deportivo.

Metodología: Se realizó una revisión de la literatura actualizada, desarrollando un artículo narrativo sobre el consumo del alcohol en la función del músculo esquelético y el rendimiento en el ejercicio, que recopila conocimientos obtenidos de estudios en humanos y modelos animales de consumo problemático del alcohol.

Resultados: Los estudios basados en deporte y ejercicio sugieren que el consumo de alcohol afecta negativamente la recuperación del músculo esquelético luego de un ejercicio vigoroso, especialmente en hombres, mientras que las mujeres se parecen menos afectadas. Por otro lado, estudios clínicos y pre-clínicos han permitido conocer algunos de los mecanismos involucrados en la disfunción muscular ocasionada por el alcohol, los que incluyen un desbalance entre las vías anabólicas y catabólicas, regeneración deficiente, un aumento en la inflamación y fibrosis, y desbalances en el equilibrio energético y la función mitocondrial. Estos mecanismos patológicos pueden ocurrir no solo en el

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consumo crónico de alcohol, sino que en otros patrones de consumo también.
Conclusiones: La mayoría de las investigaciones se han concentrado en evaluar los efectos de la exposición a alcohol aguda o crónica en el músculo esquelético, mientras que el BD, el patrón de consumo más común y frecuente en deportistas aficionados y profesionales ha sido poco estudiado. No obstante, el consumo de alcohol afecta negativamente la salud del músculo esquelético a través de mecanismos que podrían contribuir a reducir el rendimiento deportivo.

Introduction

Alcohol consumption effects on human health

The effects of alcohol consumption on human health have been the subject of extensive research. It has been claimed that moderate alcohol consumption (MAC), the occasional ingestion of up to one standard drink (14 g of pure alcohol) a day for women or two drinks a day for men (1,2), may lower the risk for some cardiovascular diseases, type-2 diabetes, dementia, and some cancers (3,4). Nevertheless, problematic alcohol consumption (PAC) prevalence is worrisome, particularly among youth and young adults (1,4). Among PAC, the National Institute on Alcohol Abuse and Alcoholism (NIAAA) from the US (1) defines two different alcohol drinking patterns, heavy drinking (HD) and binge drinking (BD), and both increase the risk of developing alcohol use disorder (1). HD is the consumption of ≥5 drinks for men or ≥4 for women on any day (in a week ≥15 for men and ≥8 for women). BD refers to the intake of large amounts of alcohol (5 drinks/men and 4 drinks/women) on a single occasion and a short time (≤2 h), reaching a blood alcohol concentration ≥ 0.08 g/dL (1,5). BD is widespread among young adults (5); for instance, a recent longitudinal study (n = 1240) of Argentinean college students revealed that only 18.8% did not exhibit BD across the 3-year follow-up, whereas 12.6% began the study exhibiting 1–2 BD per month, but declined as the study progressed. However, the remaining 70.6% of the sample reported 1–4 BD episodes per month across the three-year study (6).

According to the World Health Organization (WHO), 3.3 million deaths occur every year because of PAC (4). A significant fraction of these deaths are the consequence of a higher probability of vehicular crashes, domestic accidents, and interpersonal violence (3,4,7). Also, PAC not only increases accidents leading to traumatic injury but also worsens the progression and prognosis of the injury (8). Moreover, in the medium- and long-term, PAC is a risk factor for a plethora of health issues: malnutrition, pancreatitis, alcoholic liver diseases, cardiovascular diseases, cancer affecting the upper and lower digestive tract and other organs, and also neurodegenerative diseases and dementia (3,4,9).

Chronic alcohol is also associated with alcoholic myopathy, a debilitating condition affecting 50–60% of chronic alcohol consumers ([10–18]) characterized by skeletal muscle weakness and atrophy, predominantly of type 2 glycolytic fibers (17,18). Furthermore, PAC is also related to a greater risk of other muscle diseases, such as aging-related sarcopenia (19,20) and cirrhosis-related sarcopenia (21).

Sportspeople and the high risk for problematic alcohol consumption

Skeletal muscle corresponds to more than 40% of the body weight, is responsible for voluntary movements, regulates energy metabolism, and its contraction leads to the release of myokines, muscle-derived cytokines and chemokines acting as paracrine signals in the whole body, which are mostly responsible for the beneficial effects of exercise on health (22,23). Depending on the type, intensity, and training time, exercise modifies a plethora of physiological and psychological outcomes. For example, exercise regulates mood states, boosts the immune response, and promotes beneficial hypoglycemic effects, especially relevant to people who present morbid conditions (22). In addition, sports also engage a prominent role as a social landscape in young people offering better cognitive, physical, and social development. Thus, it is not surprising that sports practice has been a traditional approach to keep youth out of behaviors leading to PAC (24). Unfortunately, the studies that have systematically analyzed the association between sports practice and alcohol use have not fully supported this assertion. Some studies have indeed indicated a negative correlation between the variables (e.g., (25)), yet for the most part, the evidence suggests a strong positive association between the practice of sports and alcohol consumption, specifically with BD ([25–30]). Students who reported earlier sports involvement at the high school level also have a rapid increase in drinking during adolescence, and high school athletes are more likely to engage in hazardous alcohol consumption than peers who do not practice sports (26,28), albeit sport does seem to reduce the use of illicit drugs, such as

cannabis (27). Participation in college sports activities is associated with BD, particularly after a significant competition ([31–33]). Also, in the UK, the AUDIT (Alcohol Use Disorders Identification Test) scores were significantly higher in students engaged in university sports when compared to peers not reporting such engagement (34). Furthermore, many elite and amateur athletes in different countries are identified as current drinkers or problematic alcohol users, a finding particularly prominent in males and team sports (31,[35–44]).

Aim and methodology of the review

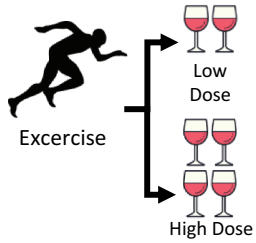
There are several unanswered questions on the relationship between sports and drinking and the effects of such drinking. One of those gaps refers to the motives underlying drinking in the context of sports. Motives for drinking involve coping with a negative situation, such as muscle soreness after a competition (45). Also, the willingness to fit into a social group (46,47), seek better social adjustment with teammates, and cope with the stress of competition (36). Another important factor contributing to alcohol use in sports environments is sponsorship: players supported by alcohol sponsors report significantly greater AUDIT scores than peers not supported by the alcohol industry (48).

The present review will focus on another facet of the alcohol–sport interaction. Amateur and professional athletes perform strenuous exercises that can produce skeletal muscle damage, inflammation, and pain, but need a rapid recovery to reach back to their optimal muscle function ([49–51]). However, as described before, they are also at increased risk of PAC. Therefore, we examined the currently available literature (available in PubMed and Google Scholar) summarizing the knowledge about the effects of alcohol on skeletal muscle function and exercise performance. We developed a narrative review that examines and discusses information obtained from studies in human beings and different animal models. We will first review seminal and more recent studies on how acute consumption, particularly in laboratory settings that allow precise control of dose, type, and magnitude of exercise, affects skeletal muscle damage and recovery after strenuous exercises. Next, we will dissect sex- and dose-related effects and potential interactions between the dose of alcohol ingested and the extent of muscle damage caused by the exercise. Then, we will focus on preclinical and clinical research assessing the cellular and molecular mechanisms underlying the effects of acute and chronic alcohol on skeletal muscle function, which can reduce muscle performance in sportspeople.

Effects of alcohol consumption on athletic performance and exercise recovery

Many athletes and sportspeople do not drink on the day of competition. However, some of them drink the day before the athletic match, following old-fashioned ideas that acute alcohol ingestion can improve their athletic performance, which has been disproved by several studies showing that alcohol impairs physical performance and cognitive function (44,[52–54]). Studies in sportspeople and team sports players suggest that alcohol is detrimental to endurance and sprint exercise performance in a dose-dependent manner ([55–57]). Additionally, alcohol ingestion the day before exercising decreases muscle performance of severe-intensity exercise (52). Also, changes in sleeping patterns associated with PAC previous to exercise can also impact on athletic performance. In young, physically active men, acute consumption of 1 g/kg of body mass alcohol combined with one night of sleep deprivation reduces quadriceps activation during maximal knee extension (58). Although the decrease was discrete (i.e., approx. 10%), the authors discuss that this can be more significant for activities that require maximum muscular effort, as in amateur and professional athletes (58). Furthermore, this study assessed the effect of just one night of alcohol and sleep deprivation, yet further detrimental effects can probably be observed if these episodes occur repetitively. Moreover, some athletes and sportspeople drink alcohol after a competition to celebrate or cope with stress (39,59,60). More recent studies analyzed the effects of post-game alcohol consumption on whole-body performance, the days following simulated rugby matches. They report detriments in muscle performance and suppression of post-match countermovement jumps, concluding that acute and heavy alcohol harms post-match recovery (39,[59–61]).

One of the first studies to establish the interaction between acute alcohol consumption and recovery from muscle damage induced by eccentric contraction (ECC) exercise was conducted in the early 1990s (62). The subjects, women aged 21–30 years, received an alcoholic beverage containing 0.8 g of ethanol per kg of body weight (0.8 g/kg) or received a nonalcoholic solution (controls). The drink was administered 35 min before performing 50 maximal ECC of the elbow flexor muscle. The study reported significant muscle damage in both groups. However, both treatments exhibited similar plasma CK activity levels, muscle soreness, and isometric strength, which concluded that this dose of alcohol did not affect the recovery of skeletal muscle performance after strenuous exercise. Since then, similar studies have been performed in men and women, yet most administered the alcohol-containing solution after the strenuous



	MEN				WOMEN			
	MUSCLE DAMAGE	TIME FOR RECOVERY	PERFOR-MANCE	CITE	DAMAGE	TIME FOR RECOVERY	PERFOR-MANCE	CITE
Low Dose	=	=	=	[64,65*]	?	?	?	[65*]
High Dose	↑ =	↑ =	↓ =	[39, 49, 59, 60, 61, 63] [65*,66*]	=	=	=	[51,67]

Figure 1. Reported effects on skeletal muscle function when alcohol has been consumed after strenuous exercise. Alcohol consumption after exercise affects muscle strength and power in a dose-dependent manner in men. In women, the evidence the evidence is limited in the amount of studies and number of subjects, especially when assessing low doses of alcohol. For example, in reference 65, only one female was evaluated. Also, training status is important, as when individuals are specifically trained for the repetitive exercise evaluated in the study (*), there is lesser muscle damage, and alcohol shows no effect on muscle recovery.

exercise. The findings of these studies are summarized in Figure 1. The first of these (49) was conducted in young men (23 ± 4 years) who drank a beverage containing 1.0 g/kg of body mass of ethanol or a nonalcoholic solution after they performed 300 maximal ECC of the quadriceps muscle. In contrast to what was found before in women, this study showed that alcohol intake enhanced muscle injury and retarded the recovery of skeletal muscle strength 24 h after performing the strenuous exercise. Furthermore, the same alcohol dose decreased maximal voluntary isometric contraction after eccentric exercise at 36 and 60 h after alcohol consumption, compared to control individuals (63). This study also focused on the effect on neuromuscular performance, suggesting that the alcohol-induced decrease in voluntary force has a neural, central, component besides the peripheral muscle damage. Interestingly, another study performed by the same research group revealed that a lower dose of alcohol (0.5 g/kg of body mass) did not affect skeletal muscle recovery from vigorous eccentric exercise (64). So then, these studies suggested that the effects of alcohol consumption on skeletal muscle recovery after exercise in men seem to depend on the ingested doses and probably the blood alcohol concentrations reached.

In contrast, later studies showed no alteration in muscle recovery after acute alcohol, neither with low (0.6 or 0.7 g/kg of body mass for females and males, respectively) nor high (1.2 or 1.4 g/kg of body mass for females and males, respectively) doses of ethanol, given 1 h after trained resistance exercise (65), or moderately high dose of alcohol (1.09 g/kg of lean mass) within 40 min of performing the same exercise task for what the subjects were previously trained (66). Interestingly, both studies discuss that the extent of muscle damage is smaller when performing a repetitive trained activity than when performing new intense muscle activities, as it was evaluated in the studies by Barnes et al. In fact, the authors informed that the extent of muscle damage observed in their samples was lower than reported before in non-trained subjects. Thus, together, these studies suggest that the effects of acute alcohol consumption on skeletal muscle recovery appear to be related to the amount of alcohol consumed (reviewed in the previous paragraph) and the extent of muscle damage caused by the exercise, which is lesser when individuals are trained for the tested task.

Two more recent studies were executed in young females, who performed 300 ECC followed by alcohol administration at doses of 0.88 (51) or 1.09 g/kg of lean mass (67). They showed that acute ingestion of the mentioned doses of alcohol did not affect skeletal muscle recovery from ECC-induced damage in females. Women from these studies were recreationally fit (51) or resistance-trained (67) but not trained for the tested exercise as in previously mentioned studies in men (65,66). Then, if considered non-trained, these studies suggest that alcohol reduces exercise recovery in men (49,63) but not in women (51,67). Interestingly, a recent study in mice focused on sex differences in skeletal muscle contractile properties after alcohol administration, showing that alcohol causes a decline in muscle force and fatigue, but only females recover after alcohol clearance (68).

Sex differences in alcohol effects on muscle function and recovery may be influenced, at least in part, due to possible sex-specific reductions in protein synthesis, which provides the basis of adaptation to resistance exercise (67,69). In humans, a study showed that alcohol consumption reduces protein synthesis by reducing mTOR phosphorylation, one of the critical signaling pathways controlling protein synthesis after resistance

exercise (70). However, these effects were not investigated in women at the time. A more recent study evaluated the effects of alcohol ingestion on mTORC1 signaling induced by resistance exercise in men and women, showing that although men and women presented similar mTORC1 signaling proteins, phosphorylation and hence the activity of this multiprotein complex was reduced only in men (69). However, the final effect on protein synthesis was not addressed, and it remains to be determined.

The hormonal role in the effects of alcohol in the regulation of skeletal muscle recovery after exercise is worth considering. Testosterone promotes muscle growth by stimulating protein synthesis and inhibiting protein degradation (71), and there is evidence indicating that alcohol consumption in men suppresses plasma levels of testosterone (72,73). However, in the context of exercise, different pieces of evidence can seem controversial. For example, a study shows that plasma levels of testosterone are not affected by high doses of alcohol (1.5 g/kg of body mass) 10 h after exercise but can reduce testosterone secretion at longer times (74). Haugvad et al. showed a dose-dependent effect of acute alcohol consumption after exercise: a low dose of ethanol (0.7 g/kg of body mass) did not alter levels of plasma testosterone, but a high dose (1.2 g/kg of body mass) decreased it (65). Nevertheless, increased testosterone levels as a consequence of alcohol consumption after exercise have also been found (75). Additionally, other parameters regulating testosterone signaling can be altered by alcohol, such as interacting proteins or androgen receptors ([75–77]). Thus, more investigation and integrative interpretation of the information is needed to understand the effects on testosterone levels and signaling given by specific doses of alcohol, training status, and its actual consequences on muscle performance.

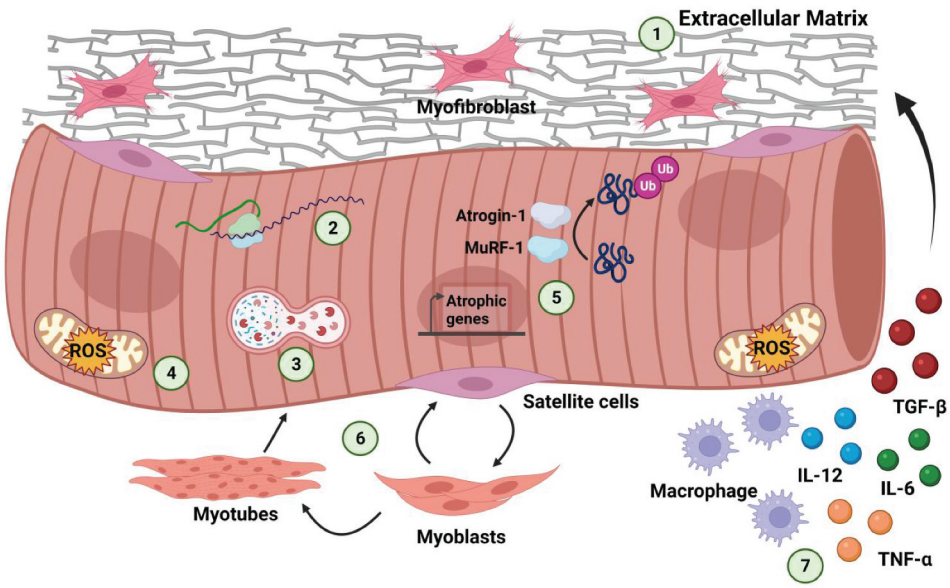
In females, animal studies also revealed a difference in muscle recovery time that could be explained by the protective effects of estrogens, which attenuates the increment of markers for muscle damage that rises after intense exercise ([78–80]). In women, increased levels of estrogens are related to some diminished markers of muscle damage after ECC (81), and reviewed evidence supports that estrogen plays an essential role in protecting skeletal muscle damage in women (82,83). Interestingly, acute ethanol boosts levels of estrogens in females ([84–86]). Although an integral view of women's health must be considered, it may be possible that elevated levels of estrogens after acute and moderate ethanol ingestion can protect women from muscle damage due to exercise, which might explain that no effects of alcohol in muscle recovery have been seen in women. Later, the decline of estrogens during menopause withdraws this protective mechanism, explaining why women present a more robust relationship between alcohol consumption and risk for sarcopenia, the aging-related muscle waste and loss of force (19,20). Nevertheless, no mechanistic studies have been performed yet to measure estrogen levels after alcohol consumption and its consequences on muscle function; then, further investigation is needed to fully understand this phenomenon.

Mechanisms underlying alcohol-related dysfunctions on skeletal muscle: what we know from clinical studies, *in vitro* research, and animal models

Besides what we can learn from patients with a history of alcohol misuse and alcoholic myopathy, laboratories have used other resources to study the cellular and molecular mechanisms that lead to alcohol-related skeletal muscle dysfunction. For example, *in vitro* experiments of skeletal myofibers exposed to acute alcohol show inhibition of intracellular Ca^{2+} transients, altering excitation–contraction coupling fundamental for muscle contraction (87). However, *in vitro* studies with isolated myofibers and differentiated myotubes have several limitations related to the complexity of the physiopathological processes involved and the interactions that actually take place between the myriad of cell types present in the tissue. Thus, animal models of alcohol effects on muscle function are required. These animal models include worms (88), mice (89), rats (90), and macaques (91), which following alcohol exposure exhibit some of the features seen in humans, including striated muscle atrophy, decreased muscle contractility, and increased muscle fatigue. The main findings regarding molecular changes in skeletal muscle due to alcohol are reviewed in this chapter and summarized in Figure 2.

Alcohol alters the balance of signaling pathways that maintain muscle mass

The skeletal muscle mass is highly coordinated and regulated by different signaling pathways, extensively studied and reviewed (92,93), whose imbalance can lead to muscle hypertrophy or atrophy. Insulin-like growth factor (IGF-1) is an anabolic factor that promotes muscle growth, improves regeneration, and maintains glucose homeostasis. Upon binding to its receptor, IGF-1 activates PI3K/Akt/mTOR (92). The mTOR signaling pathway is a crucial regulator of muscle mass, mediating the phosphorylation of downstream factors that promotes protein synthesis by activating ribosomal protein S6 and releasing the translation initiation factor eIF-4E (94). In addition, catabolic pathways are stimulated by growth factors and



	Process	Effect	Markers
Increased	Inflammation and fibrosis	Inflammatory response and deposition of connective tissue.	↑TNF-α [121], ↑IL-6 [121], ↑IL-12 [121], ↑TGF-β1 [98, 111, 112, 122] ↑collagen [111, 121, 122], ↑fibronectin [111, 122], ↑α-smooth muscle actin [121].
	Oxidative stress	Mitochondrial dysfunction and oxidative stress	↓NAD+/NADH ratio [101], ↑intermediates of citric acid cycle [101], ↑mitochondrial UPR response [146], ↑lipid and protein oxidation [101, 111], ↓antioxidant potential [99, 101, 111].
	Catabolism	Augmented protein degradation	↑atrogin-1* [97-99], ↑MuRF-1* [97-99] ↑ Autophagy [100-102].
Decreased	Anabolism	Reduced protein synthesis	↓IGF-1 [94], ↓p-Akt [94], ↓p-4E-BP1 [91-94], ↓mTOR signaling * [91-93].
	Muscle regeneration	Delay in myoblast proliferation and differentiation.	↓CK in differentiation [108, 109] ↑Notch signaling [110] ↓MyoD [110, 112-115], ↓myogenin [110, 113-115], ↓MEF2C [112-115].

Figure 2. Cellular and molecular alterations in problematic drinking that lead to muscle wasting. Skeletal muscle damage associated with heavy and chronic alcohol consumption is characterized by an imbalance in mechanisms and signaling pathways that regulates anabolism/catabolism, decreasing muscle mass: decreased protein translation (2), enhanced protein degradation (5), and increased autophagy (3). Other molecular features of alcohol-related myopathy are increased inflammation (7), fibrotic response and exacerbated extracellular matrix accumulation (1), and mitochondrial dysfunction with increased oxidative stress (4). Also, the reduced regeneration capacity of the muscle (6) has been described, although most the research in this field come from studies in immunodeficient animals, and more research is needed to determine if the same alcohol effects are observed in healthy individuals. In the table, we summarize for each process the main molecular markers that have been found increased (↑) or decreased (↓) in response to alcohol, indicating in brackets the published references. (*) indicate molecular markers that have been found no changed in some studies in females. This figure was created with BioRender.com.

inflammatory cytokines such as myostatin, TNF-α, and IL-6, among others, leading to the induction of transcription factors such as NF-κB and FoxO 3, which results in the expression of E3 ligases MuRF-1 and MAFbx/Atrogin-1 with consequent protein degradation by the ubiquitin-proteasome system (UPS) (92).

It has been described that both acute intoxication and chronic alcohol abuse decrease basal levels of protein synthesis by mechanisms depending on mTOR signaling in animal models ([95–97]). Despite the decrease of the total amount of skeletal muscle proteins, the downregulation of specific proteins of the contractile apparatus, such as titin and nebulin, have been found in samples from women with a history of chronic alcohol misuse (98) and in rats exposed to chronic alcohol consumption (approx. 24 g/kg/day) (90,99). In humans, proteins from anabolic pathways, such as IGF-1, p-Akt, and p-4E-BP1, are downregulated in chronic alcohol consumers compared with the control group, suggesting decreased protein translation (98). Acute alcohol administration in male rats also increases atrogin-1 and Murf-1 mRNA (no protein levels evaluated) without an immediate increase in proteolysis (100), which might occur later in response to repetitive alcohol exposure. Indeed, skeletal muscle atrophy in male rats after chronic alcohol consumption is preceded by increased atrogin-1 and Murf-1 gene expression, which returns to baseline levels when atrophy is established (101,102). Increased catabolism also appears to be related to increased autophagy, observed *in vitro, in vivo* in mouse models, and skeletal muscle biopsies from patients with chronic alcohol use (89,103,104). Interestingly, studies in female mice show that alcohol reduces protein synthesis, independently of mTOR signaling, but without changes in catabolic pathways such as in mRNA levels of atrogin-1 and Murf-1 or autophagy markers (105). This difference might help maintain the balance between anabolic/catabolic pathways and contribute to sex differences in alcohol effects on muscle function discussed before. Furthermore, and intriguingly, local intracerebroventricular (ICV) ethanol infusion reversibly decreases protein synthesis by increasing muscle proteasome activity. This study reveals a role for alcohol effects on the central nervous system affecting skeletal muscle protein homeostasis independently of the effects of systemic alcohol circulation (106).

Remarkably, alterations in the balance between anabolic and catabolic pathways in alcohol-related disorders can be observed even before the appearance of evident skeletal muscle atrophy, any evident force loss, or other clinical symptoms for alcohol-related myopathy (101,107), and might be contributing to decreasing skeletal muscle performance in sportspeople with problematic drinking.

Alcohol reduces the regeneration capacity of skeletal muscle

Decreased muscle performance and slow recovery can also be caused by altered regeneration after muscle damage (108,109). Satellite cells and regeneration are involved in muscle growth in muscle overload and exercise-induced hypertrophy (110,111), and dysfunction in the regeneration process is involved in muscle waste pathologies such as sarcopenia (112,113). Furthermore, skeletal muscle damage is recurrent in athletes, from micro to macro injuries, resulting from accidents, overuse, and overtraining ([114–116]); then, possible defective regeneration due to PAC can contribute to poor recovery and decreased performance.

In vitro experiments with rat primary myoblasts exposed to ethanol in the culture media show that alcohol-exposed myoblasts proliferate less than controls, and the expected increment in creatine kinase (CK) during differentiation is inhibited (117,118). In addition, the formation of C2C12 myotubes is delayed by alcohol, decreasing the expression of transcription factors MyoD and myogenin, which are early markers for differentiation, and concomitant abnormal activation of the Notch signaling pathway, which is known to inhibit myoblast differentiation (119). *In vivo*, chronic alcohol consumption in mice (ad libitum exposure to 20% v/v alcohol, for circa 5 months) leads to reduced fiber caliber in regenerated fibers 14 days after $BaCl_2$ damage in tibial anterior (TA) muscle (120). However, in the same study, some atrophy was also observed in the contralateral, not damaged, TA, evidencing an intrinsic atrophic effect of chronic alcohol, exacerbated in the damaged hindlimb. This intrinsic hindlimb atrophy is also observed in rats exposed to chronic alcohol (90,99).

In rats, chronic alcohol also caused the reduction of regeneration markers Mef2c and MyoD early after recovery (3 days) from immobilization (121). Furthermore, immunodeficient Rhesus macaques exposed to a chronic binge alcohol protocol displayed exacerbated muscle wasting (122), which might have been caused by reduced regeneration, as the differentiation of primary myoblasts is inhibited in samples from macaques exposed to alcohol (123). The mechanisms for this observation pointed to the reduction of *PAX7, MYOD1, MYOG, MYF5*, and *MEF2C* expression, which are all necessary for myoblast differentiation, and to decreased glycolytic metabolic function (91,122,123). Therefore, studies *in vitro and in vivo* suggest that alcohol exposure can delay normal myoblast proliferation and differentiation, which might affect recovery in athletes.

Alcohol induces inflammation and fibrotic response in skeletal muscle

Acute inflammation is required for proper skeletal muscle regeneration after injury. However, persistent damage and inflammation can lead to fibrosis, a pathological feature that forms a non-contractile, scar-like tissue caused by increased deposition of fibrillar

collagen and other extracellular matrix (ECM) proteins, such as fibronectin and proteoglycans (124,125). Profibrotic factors such as TGF-β and CTGF/CCN2 are increased and modulate fibrosis in different pathologies, for example, muscular dystrophies, chronic muscle damage, denervation, ALS, and muscle overuse; also, decreasing fibrosis by inhibition of those factors can improve muscle function ([126–132]). The establishment of fibrosis can affect skeletal muscle force production, impair regeneration and reinnervation, decrease oxygen availability, among others (131,133,134) and has been discussed in the context of sports injury ([135–137]).

PAC leads to abnormally elevated inflammation in skeletal muscle, manifested by increased levels of TNF-α, IL-6, IL-12, and other proinflammatory cytokines (120,121,138,139). TGF-β1 mRNA and/or protein is increased in skeletal muscle from alcohol-fed mice, rats, and macaques (101,120,121,139). Also, ECM proteins collagen and fibronectin levels are significantly increased in injured skeletal muscle from chronic alcohol-fed mice (120). Nevertheless, TGF-β and ECM proteins are also elevated in the uninjured tibial anterior compared to control-fed mice, suggesting that chronic alcohol consumption can also intrinsically induce fibrosis in the non-damaged muscle (120). Increased fibrosis is also observed in chronically alcohol-fed rats, which present elevated mRNA or protein levels for many ECM components and increased α-smooth muscle actin protein, a marker for activated myofibroblasts, cells that produce excessive amounts of ECM (138). In immunosuppressed Rhesus macaques, fibrosis is also suggested upon chronic binge alcohol, which leads to the upregulation of ECM remodeling genes and increased collagen deposition (139). Therefore, we suggest that PAC can increase inflammation and fibrosis in sportspeople, contributing to decreased muscle performance for reasons previously exposed, aggravating the consequences of muscle damage during exercise and tournaments.

Alcohol leads to altered metabolism, mitochondrial dysfunction, and increased oxidative stress in skeletal muscle

Changes in glucose metabolism induced by alcohol have been associated with decreased muscle performance in athletes, including impaired gluconeogenesis and glucose output, decreased lactate and glycerol (140,141), and reduced glucose uptake and glycogen storage (142). Recent studies, however, indicate that the use of glycogen in exercise is similar with or without consuming alcohol (143). Early studies showed that muscle damage due to chronic alcohol exposure was not related to mitochondrial metabolism imbalance in rats (oxygen uptake and ATPse activity) (144) and humans (oxygen uptake, cytochrome content, respiratory chain complexes) (145). However, more recent analyses have established a strong relationship between skeletal muscle damage and dysregulated mitochondrial metabolism because of alcohol consumption. *In vitro* experiments and *in vivo* studies in mouse models have shown that alcohol alters basal energetic metabolism by reducing NAD^+/NADH ratio, induces accumulation of intermediates of the tricarboxylic acid cycle, and induces malfunction of the respiratory chain (89).

Mitochondrial health, mainly maintained by mitochondrial fusion/fission, is necessary for maintaining energy homeostasis and appropriate Ca^{2+} signaling, crucial for skeletal muscle contraction. Myofibers isolated from chronically alcohol-fed rats (36% of total calories, for 6–11 months) present Ca^{2+} dysregulation due to decreased mitochondrial Ca^{2+} uptake, associated with decreased mitochondrial fusion and reduction of Mfn-1 protein, which is necessary for that process (146). Later, decreased ATP and oxygen consumption, evidencing mitochondrial dysfunction, was found in the skeletal muscle of patients from alcoholic cirrhosis, cultured myotubes exposed to alcohol, and mice and Rhesus macaques exposed to chronic alcohol intake (89,[147–149]). Also, in *C. elegans*, increased oxidative stress upon alcohol exposure is accompanied by increased fragmentation of the mitochondrial network and increased mitochondrial unfolded protein response (UPR^{mt}) (88). Although a normal defense response to stress, like inflammation, UPR^{mt} can be deleterious when persisting in time, allowing the spread of mitochondrial damage by avoiding mitophagy, a process that degrades defective mitochondria (150). In this respect, the absence of Parkin, an E3 ligase protein that facilitates mitophagy, exacerbates chronic alcohol-induced injury in skeletal muscle, including mitophagy dysfunction (which also occurs in the liver and brain) (149,151). Recessive mutations of Parkin are linked to Parkinson's disease, and interestingly, when Parkin is absent, the mitochondrial and contractile functions of muscle are altered, evidencing a critical role in the maintenance of skeletal muscle structure and function (151). Thus, like with Parkin, alcohol-deleterious effects might be exacerbated on specific genetic backgrounds, increasing the risks for some chronic and degenerative diseases or decreasing exercise performance.

Oxidative stress, which can be a cause or consequence of mitochondrial dysfunction, also appears to be a marker for alcohol-induced skeletal muscle damage. In mice and rats, chronic alcohol-induced skeletal muscle oxidative stress and decreased antioxidant potential are evidenced by changes in Glutathione (GSH), oxidized glutathione (GSSG), augmented lipid peroxidation, and increased total protein oxidation (89,102,120). Furthermore, oxidative stress measured as glutathione (GSH), oxidized

glutathione (GSSG), and total protein oxidation was more abundant during regeneration of $BaCl_2$-injured TA of alcohol-fed than from control diet rats (120). Altogether, these data demonstrate that oxidative stress and mitochondrial dysfunction can delay regeneration and recovery of muscle function and contribute to alcohol-related dysfunctions.

Summary and perspectives

This work summarizes the effects of alcohol consumption on skeletal muscle function and how they might contribute to decreased muscle performance in sportspeople. This analysis is meaningful in light of the reports of a positive association between the practice of sports and problematic alcohol consumption ([25–28]).

It is essential to highlight that several other alcohol effects could affect athletic performance in competition, tournament, or training. Those effects are not reviewed here but include alterations of diuretic, vasodilator, and aerobic capacity, changes in systemic glucose metabolism, effects on cognitive function that reduces decision-making speed, quality of response to visual stimuli, and the ability to control instruments (44,54,59,143,152). Furthermore, it is essential to remark that caution should be exercised when trying to understand and extrapolate some of the reviewed effects of PAC on the daily, regular sports practice. Limitations of the studies in humans reviewed here include the variability of physical demands and training status, the lack of contact components during simulated matches (59), the fact that the dose of administration in human studies can be considerably lower than the volume of alcohol actually consumed by athletes (60), and the possible lack of control of the actual amounts and number of events of alcohol ingestion by elite sportspeople when using categorized questionaries for self-report alcohol use (39,61).

Clinical studies and laboratory research have led to the knowledge of some of the mechanisms involved in alcohol-related muscle features. These mechanisms, shared with well-studied muscle diseases, include an imbalance in anabolic and catabolic pathways leading to muscle atrophy, impaired regeneration, increased inflammation, the establishment of a fibrotic process, and poor mitochondrial metabolism with increased oxidative stress, and could worsen the prognosis for recovery after damage in sportspeople. However, most laboratory-based studies use chronic or acute (one dose) alcohol exposure, while BD, the most common drinking pattern in college and professional athletes (31,32,153), is underrepresented. BD has also been studied in the background of other severe conditions, such as immunodeficiency (122,123,139,148,154), which can show other effects not necessarily found in healthy sport-related individuals. Furthermore, a study in mice concluded that only chronic alcohol consumption (30% of calorie intake for 14 weeks), but not binge alcohol (3 g/kg), decreases contractile function measured as normalized tetanic force and fatigue in isolated EDL muscle (155). However, the binge protocol consisted of a single exposure, and contractile function was measured only 2 h after alcohol administration. Then, molecular alterations in response to alcohol that take longer to develop would be missed from the observations. Instead, BD in humans, especially during youth and sport-related people, corresponds to episodic repetitive events ([31–33,]153'156), which was not addressed in the mentioned study. Thus, although significant advances have been made in the field, more research is needed to relate different relevant drinking patterns, especially BD, to specific muscle pathological features that can contribute to decrease muscle performance in amateur and professional athletes.

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